task1.py × task1_output.txt

```python
import os
import platform
import time
import numpy as np

STUDENT_ID = "230103227"
ARRAY_SIZE = 20_000_000

print("Student ID:", STUDENT_ID)
print("Operating System:", platform.platform())
print("CPU:", platform.processor() or platform.machine())
print("Logical CPU cores:", os.cpu_count())

rng = np.random.default_rng(int(STUDENT_ID[-3:]))
data = rng.random(ARRAY_SIZE, dtype=np.float64)

start = time.perf_counter()

total_sum = 0.0
for value in data:
    total_sum += value

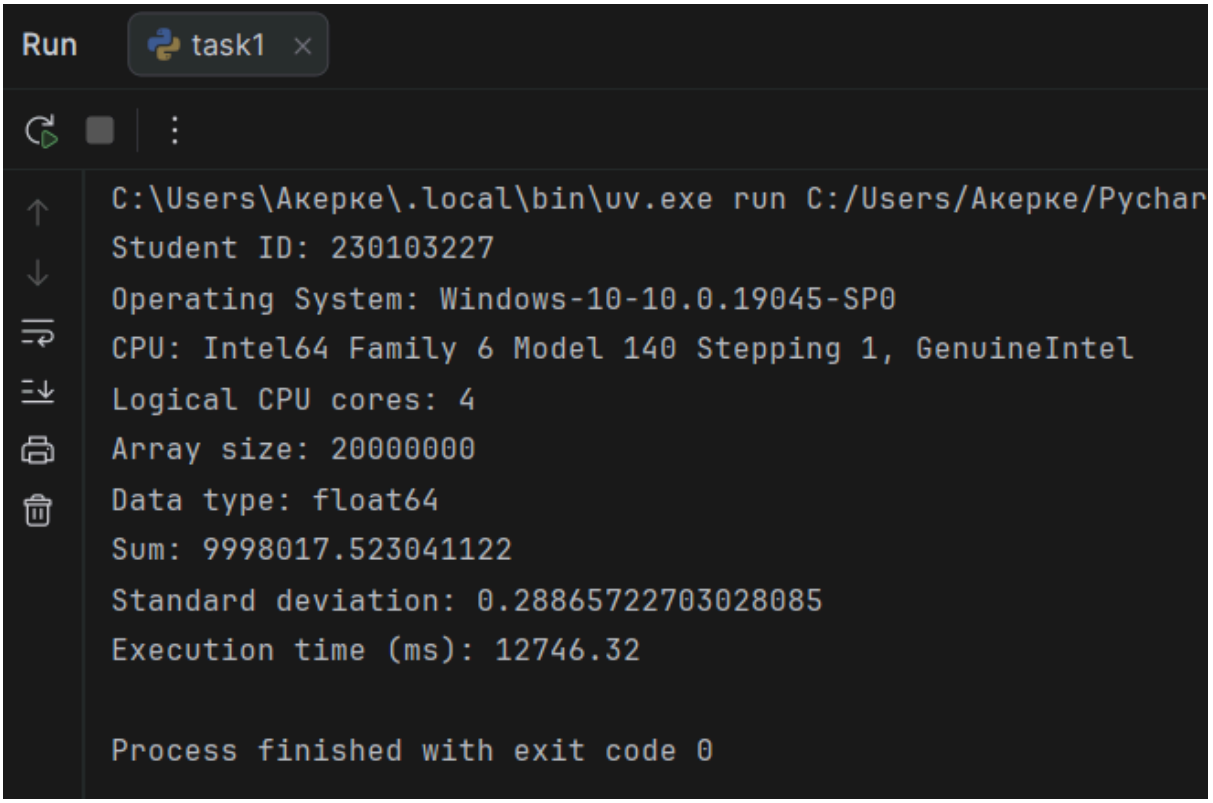
mean = total_sum / ARRAY_SIZE

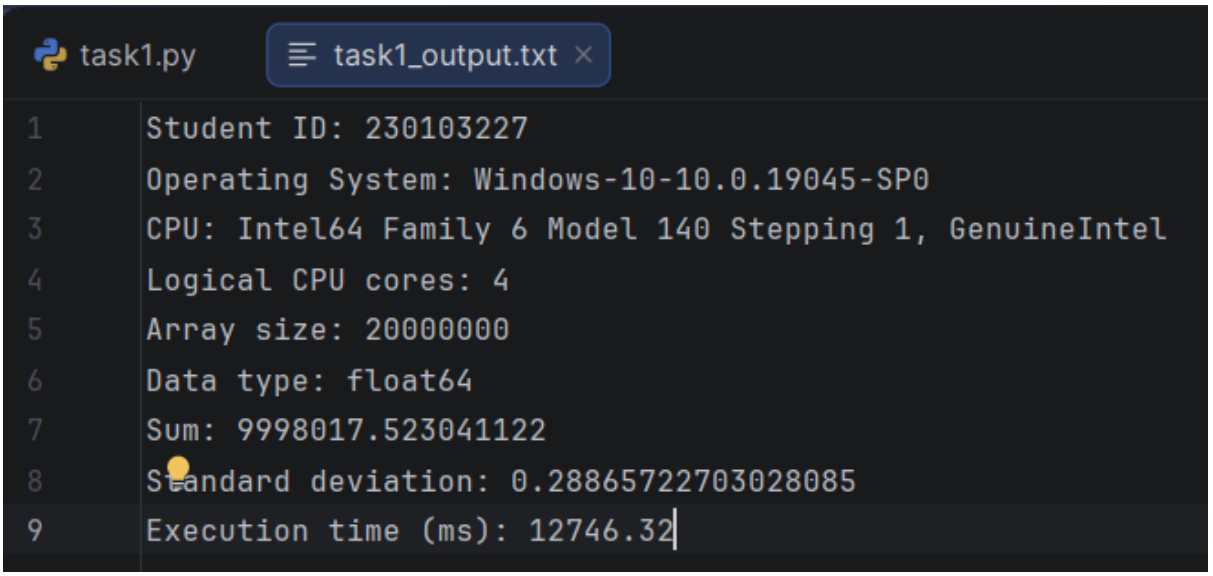
squared_diff_sum = 0.0
for value in data:
    diff = value - mean
    squared_diff_sum += diff * diff

std_dev = (squared_diff_sum / ARRAY_SIZE) ** 0.5

end = time.perf_counter()

print("Array size:", ARRAY_SIZE)
print("Data type:", data.dtype)
print("Sum:", total_sum)
print("Standard deviation:", std_dev)
print("Execution time (ms):", round((end - start) * 1000, 2))
```


Run
task1
C:\Users\Акерке\.local\bin\uv.exe run C:/Users/Акерке/Pychar
Student ID: 230103227
Operating System: Windows-10-10.0.19045-SP0
CPU: Intel64 Family 6 Model 140 Stepping 1, GenuineIntel
Logical CPU cores: 4
Array size: 20000000
Data type: float64
Sum: 9998017.523041122
Standard deviation: 0.28865722703028085
Execution time (ms): 12746.32
Process finished with exit code 0


task1.py
task1_output.txt
1 Student ID: 230103227
2 Operating System: Windows-10-10.0.19045-SP0
3 CPU: Intel64 Family 6 Model 140 Stepping 1, GenuineIntel
4 Logical CPU cores: 4
5 Array size: 20000000
6 Data type: float64
7 Sum: 9998017.523041122
8 Standard deviation: 0.28865722703028085
9 Execution time (ms): 12746.32